

Lecture 1: Inequalities

1. Introduction

An **inequality** compares two real numbers or expressions using the symbols

$$<, >, \leq, \geq.$$

Many optimization problems and proof-type questions can be solved by applying standard inequalities.

In this lecture we study

- Arithmetic Mean (AM)
 - Geometric Mean (GM)
 - Harmonic Mean (HM)
 - Weighted Mean Inequality
 - Cauchy Inequality
 - Weierstrass Inequality
 - Chebyshev Inequality
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2. Arithmetic, Geometric and Harmonic Mean

Suppose

$$a_1, a_2, \dots, a_n > 0.$$

Arithmetic Mean (AM)

$$AM = \frac{a_1 + a_2 + \dots + a_n}{n}.$$

Geometric Mean (GM)

$$GM = \sqrt[n]{a_1 a_2 \dots a_n}.$$

Harmonic Mean (HM)

$$HM = \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}}.$$

AM–GM–HM Inequality

For positive numbers,

$$\boxed{AM \geq GM \geq HM.}$$

Equality holds iff

$$a_1 = a_2 = \dots = a_n.$$

Arithmetic Mean $\geq$ Geometric Mean

Theorem

If $a, b > 0$, then

$$\frac{a + b}{2} \geq \sqrt{ab}.$$

Equality holds iff $a = b$.

Proof

Since every square is nonnegative,

$$(a - b)^2 \geq 0.$$

Expanding,

$$a^2 - 2ab + b^2 \geq 0.$$

Adding $4ab$ to both sides,

$$a^2 + 2ab + b^2 \geq 4ab.$$

Hence,

$$(a + b)^2 \geq 4ab.$$

Taking positive square roots,

$$a + b \geq 2\sqrt{ab}.$$

Therefore,

$$\frac{a + b}{2} \geq \sqrt{ab}.$$

Equality holds iff

$$(a - b)^2 = 0,$$

i.e.,

$$a = b.$$

Three-variable form

$$\frac{a + b + c}{3} \geq \sqrt[3]{abc}.$$

Example 1

Show that

$$a + \frac{1}{a} \geq 2.$$

Solution

Apply AM-GM to

$$a, \frac{1}{a}.$$

Then

$$\frac{a + \frac{1}{a}}{2} \geq \sqrt{a \cdot \frac{1}{a}} = 1.$$

Hence

$$\boxed{a + \frac{1}{a} \geq 2.}$$

Equality holds when

$$a = 1.$$

Example 2

If

$$x + y = 12,$$

find the maximum value of

$$xy.$$

Solution

By AM-GM,

$$\frac{x + y}{2} \geq \sqrt{xy}.$$

Therefore,

$$6 \geq \sqrt{xy}$$

so

$$xy \leq 36.$$

Hence

$$\boxed{\max(xy) = 36.}$$

Equality occurs when

$$x = y = 6.$$

Example: Prove $\left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2 \geq \frac{25}{2}, a + b = 1, a, b > 0$.

Solution: Expand the left-hand side:

$$\begin{aligned} L &= \left(a^2 + 2 + \frac{1}{a^2}\right) + \left(b^2 + 2 + \frac{1}{b^2}\right) \\ &= a^2 + b^2 + 4 + \frac{1}{a^2} + \frac{1}{b^2}. \end{aligned}$$

Since, $a + b = 1$, we have $a^2 + b^2 = (a + b)^2 - 2ab = 1 - 2ab$.

Hence, $L = 5 - 2ab + \frac{1}{a^2} + \frac{1}{b^2}$. Now apply AM-GM: $\frac{1}{a^2} + \frac{1}{b^2} \geq \frac{2}{ab}$. Therefore, $L \geq 5 - 2ab + \frac{2}{ab}$.

Let, $t = ab$. Since $a + b = 1, 0 < t \leq \frac{1}{4}$.

Define $f(t) = 5 - 2t + \frac{2}{t}$. Now $f'(t) = -2 - \frac{2}{t^2} < 0$, so $f(t)$ is decreasing.

Since $t \leq \frac{1}{4}, f(t) \geq f\left(\frac{1}{4}\right)$.

Thus

$$L \geq 5 - \frac{1}{2} + 8 = \frac{25}{2}.$$

Hence

$$\boxed{\left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2 \geq \frac{25}{2}}.$$

Equality holds when

$$a = b = \frac{1}{2}.$$

DIY: Find the greatest value of x^3y^4 subject to $3x + 2y = 3, x, y > 0$.

AM $\geq$ GM $\geq$ HM

Proof

From AM $\geq$ GM,

$$\frac{a + b}{2} \geq \sqrt{ab}.$$

Replace

$$a \rightarrow \frac{1}{a}, b \rightarrow \frac{1}{b}.$$

Then

$$\frac{\frac{1}{a} + \frac{1}{b}}{2} \geq \frac{1}{\sqrt{ab}}.$$

Taking reciprocals,

$$\frac{2ab}{a+b} \leq \sqrt{ab}.$$

But

$$HM = \frac{2ab}{a+b}.$$

Hence

$$\boxed{AM \geq GM \geq HM.}$$

3. Weighted Mean Inequality

Theorem

Suppose

$$a, b > 0$$

and

$$m, n > 0,$$

then

$$\boxed{\frac{ma + nb}{m + n} \geq a^{\frac{m}{m+n}} b^{\frac{n}{m+n}}.}$$

Proof

Suppose

$$m, n$$

are positive integers.

Write

$$ma + nb = \underbrace{a + \cdots + a}_m + \underbrace{b + \cdots + b}_n.$$

There are

$$m + n$$

numbers.

Applying AM-GM,

$$\frac{ma + nb}{m + n} \geq \sqrt[m+n]{a^m b^n}.$$

Hence,

$$\boxed{\frac{ma + nb}{m + n} \geq a^{\frac{m}{m+n}} b^{\frac{n}{m+n}}}.$$

Equality holds iff

$$a = b.$$

General Form

If

$$\lambda_1 + \lambda_2 + \cdots + \lambda_n = 1,$$

where

$$\lambda_i > 0,$$

then

$$\boxed{\lambda_1 x_1 + \cdots + \lambda_n x_n \geq x_1^{\lambda_1} x_2^{\lambda_2} \cdots x_n^{\lambda_n}}.$$

Application

Weighted AM-GM is mainly used for

- optimization problems
- maximum/minimum value problems

Example 3

Find the greatest value of

$$x^2 y^3$$

subject to

$$4x + 3y = 12.$$

Solution

Rewrite

$$4x + 3y = 2x + 2x + y + y + y.$$

There are

$$5$$

positive terms.

Applying AM-GM,

$$\frac{4x + 3y}{5} = \frac{2x + 2x + y + y + y}{5} \geq \sqrt[5]{(2x)(2x)yyy}.$$

$$\frac{12}{5} \geq \sqrt[5]{(2x)(2x)yyy}.$$

Therefore

$$\left(\frac{12}{5}\right)^5 \geq 4x^2y^3.$$

Hence

$$x^2y^3 \leq \frac{1}{4}\left(\frac{12}{5}\right)^5.$$

Therefore

$$\boxed{\max(x^2y^3) = \frac{1}{4}\left(\frac{12}{5}\right)^5.}$$

Equality occurs when

$$2x = y = \frac{12}{5}.$$

Hence

$$\boxed{x = \frac{6}{5}, y = \frac{12}{5}.}$$

Example 4 (Exam Type)

Find the greatest value of

$$x^3y^4$$

subject to

$$3x + 4y = 3.$$

Exercise.

(Hint: Write

$$3x + 4y = x + x + x + y + y + y + y.$$

)

4. Cauchy Inequality

Statement

For real numbers,

$$a_1, a_2, \dots, a_n,$$

and

$$b_1, b_2, \dots, b_n,$$
$$(a_1^2 + \dots + a_n^2)(b_1^2 + \dots + b_n^2) \geq (a_1 b_1 + \dots + a_n b_n)^2.$$

Equality iff

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \dots = \frac{a_n}{b_n}.$$

Proof

Consider

$$\sum_{i=1}^n (a_i t - b_i)^2 \geq 0.$$

Expanding,

$$t^2 \sum a_i^2 - 2t \sum a_i b_i + \sum b_i^2 \geq 0.$$

This quadratic is nonnegative for every

t.

Therefore,

its discriminant is nonpositive.

Hence,

$$4(\sum a_i b_i)^2 - 4(\sum a_i^2)(\sum b_i^2) \leq 0.$$

Thus,

$$(\sum a_i b_i)^2 \leq (\sum a_i^2)(\sum b_i^2).$$

Equality holds iff

$$a_i = \lambda b_i.$$

Important Special Case

Take

$$a_1 = x, a_2 = y, a_3 = z, \text{ and } b_i = 1.$$

Then

$$(x + y + z)^2 \leq 3(x^2 + y^2 + z^2).$$

Example 5

If

$$x^2 + y^2 + z^2 = 27,$$

show that

$$x + y + z \leq 9.$$

Solution

By Cauchy,

$$(x + y + z)^2 \leq 3(x^2 + y^2 + z^2) = 81.$$

Hence

$$x + y + z \leq 9.$$

Example:

If x, y, z are positive real numbers such that $x^2 + y^2 + z^2 = 27$, show that $x^3 + y^3 + z^3 \geq 81$, using the Cauchy–Schwarz inequality.

Start with the Cauchy–Schwarz inequality in the form

$$\left(\sum a_i^2 \right) \left(\sum b_i^2 \right) \geq \left(\sum a_i b_i \right)^2.$$

Choose

$$a_i = \sqrt{x_i}, b_i = x_i^{3/2},$$

where $x_1 = x, x_2 = y, x_3 = z$.

Then

$$\begin{aligned} \sum a_i^2 &= x + y + z, \\ \sum b_i^2 &= x^3 + y^3 + z^3, \end{aligned}$$

and

$$\sum a_i b_i = x^2 + y^2 + z^2.$$

Therefore, Cauchy–Schwarz gives

$$(x + y + z)(x^3 + y^3 + z^3) \geq (x^2 + y^2 + z^2)^2.$$

Hence,

$$(x + y + z)(x^3 + y^3 + z^3) \geq 27^2 = 729.$$

Again, by the Cauchy–Schwarz inequality,

$$(x + y + z)^2 \leq 3(x^2 + y^2 + z^2) = 81.$$

Therefore,

$$x + y + z \leq 9.$$

Using this in the previous inequality,

$$9(x^3 + y^3 + z^3) \geq 729,$$

which implies

$$x^3 + y^3 + z^3 \geq 81.$$

Hence,

$$\boxed{x^3 + y^3 + z^3 \geq 81.}$$

5. Weierstrass Inequality

Statement

If

$$a_i > -1,$$

then

$$\boxed{(1 + a_1)(1 + a_2) \cdots (1 + a_n) \geq 1 + a_1 + a_2 + \cdots + a_n.}$$

Equality holds if at most one

$$a_i$$

is nonzero.

Proof

We prove by induction.

For

$$\begin{aligned} n &= 2, \\ (1 + a)(1 + b) &= 1 + a + b + ab. \end{aligned}$$

Since

$$\begin{aligned} ab &\geq 0, \\ (1 + a)(1 + b) &\geq 1 + a + b. \end{aligned}$$

hence, the proof.

Application

Useful for proving product inequalities.

6. Chebyshev Inequality

Suppose

$$a_1 \leq a_2 \leq \cdots \leq a_n$$

and

$$b_1 \leq b_2 \leq \cdots \leq b_n.$$

Then

$$\boxed{\frac{1}{n} \sum a_i b_i \geq \left(\frac{1}{n} \sum a_i \right) \left(\frac{1}{n} \sum b_i \right)}.$$

If the sequences are oppositely ordered,
the inequality reverses.

Example 1

If a, b, c are positive integers and the sum of any two is not greater than the third, prove that

$$\left(1 + \frac{b-c}{a}\right)^a \left(1 + \frac{c-a}{b}\right)^b \left(1 + \frac{a-b}{c}\right)^c \leq 1.$$

Solution:

First simplify each factor.

Observe that

$$\begin{aligned} 1 + \frac{b-c}{a} &= \frac{a+b-c}{a}, \\ 1 + \frac{c-a}{b} &= \frac{b+c-a}{b}, \\ 1 + \frac{a-b}{c} &= \frac{c+a-b}{c}. \end{aligned}$$

Hence

$$L = \left(\frac{a+b-c}{a} \right)^a \left(\frac{b+c-a}{b} \right)^b \left(\frac{c+a-b}{c} \right)^c.$$

Apply the **Weighted AM–GM Inequality**

$$\left(\frac{x_1}{\lambda_1} \right)^{\lambda_1} \left(\frac{x_2}{\lambda_2} \right)^{\lambda_2} \left(\frac{x_3}{\lambda_3} \right)^{\lambda_3} \leq \left(\frac{x_1 + x_2 + x_3}{\lambda_1 + \lambda_2 + \lambda_3} \right)^{\lambda_1 + \lambda_2 + \lambda_3},$$

where

$$x_1 = a + b - c, x_2 = b + c - a, x_3 = c + a - b,$$

and

$$\lambda_1 = a, \lambda_2 = b, \lambda_3 = c.$$

Since

$$(a + b - c) + (b + c - a) + (c + a - b) = a + b + c,$$

we obtain

$$L \leq \left(\frac{a + b + c}{a + b + c} \right)^{a+b+c} = 1.$$

Hence

$$\left(1 + \frac{b-c}{a}\right)^a \left(1 + \frac{c-a}{b}\right)^b \left(1 + \frac{a-b}{c}\right)^c \leq 1.$$

Example 2

Show that

$$a^8 + b^8 + c^8 > a^3 b^3 c^3 \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right),$$

where a, b, c are unequal positive numbers.

Solution:

Observe that

$$a^3 b^3 c^3 \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) = a^2 b^3 c^3 + a^3 b^2 c^3 + a^3 b^3 c^2.$$

Therefore it is enough to prove

$$a^8 + b^8 + c^8 \geq a^2 b^3 c^3 + a^3 b^2 c^3 + a^3 b^3 c^2.$$

Now apply AM–GM.

For the first term,

$$\frac{2a^8 + 3b^8 + 3c^8}{8} \geq \sqrt[8]{a^{16} b^{24} c^{24}} = a^2 b^3 c^3.$$

Hence

$$2a^8 + 3b^8 + 3c^8 \geq 8a^2 b^3 c^3.$$

Similarly,

$$3a^8 + 2b^8 + 3c^8 \geq 8a^3 b^2 c^3,$$

and

$$3a^8 + 3b^8 + 2c^8 \geq 8a^3 b^3 c^2.$$

Adding the three inequalities,

$$8(a^8 + b^8 + c^8) \geq 8(a^2 b^3 c^3 + a^3 b^2 c^3 + a^3 b^3 c^2).$$

Therefore,

$$a^8 + b^8 + c^8 \geq a^2 b^3 c^3 + a^3 b^2 c^3 + a^3 b^3 c^2.$$

Since

$$a, b, c$$

are **unequal**, equality in AM–GM cannot occur. Hence the inequality is strict:

$$a^8 + b^8 + c^8 > a^3 b^3 c^3 \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right).$$

MORE EXAMPLES:

Part A: AM–GM–HM Inequality

Question 1

Show that

$$a^6 + b^6 \geq 2a^3 b^3.$$

Solution

By AM–GM,

$$\frac{a^6 + b^6}{2} \geq \sqrt{a^6 b^6} = a^3 b^3.$$

Hence,

$$a^6 + b^6 \geq 2a^3 b^3.$$

Equality holds when

$$a = b.$$

Question 2

Show that

$$a^4 + b^4 + c^4 \geq abc(a + b + c).$$

Solution

Apply AM–GM separately.

$$\frac{a^4 + a^4 + b^4 + c^4}{4} \geq \sqrt[4]{a^8 b^4 c^4} = a^2 bc.$$

Hence,

$$2a^4 + b^4 + c^4 \geq 4a^2 bc.$$

Similarly,

$$\begin{aligned} a^4 + 2b^4 + c^4 &\geq 4ab^2 c, \\ a^4 + b^4 + 2c^4 &\geq 4abc^2. \end{aligned}$$

Adding,

$$4(a^4 + b^4 + c^4) \geq 4abc(a + b + c).$$

Therefore,

$$a^4 + b^4 + c^4 \geq abc(a + b + c).$$

Question 3

Show that

$$a + \frac{4}{a} \geq 4.$$

Solution

By AM–GM,

$$\frac{a + \frac{4}{a}}{2} \geq \sqrt{a \cdot \frac{4}{a}} = 2.$$

Hence,

$$\boxed{a + \frac{4}{a} \geq 4.}$$

Equality holds when

$$a = 2.$$

Question 4

Show that

$$\frac{1}{a} + \frac{1}{b} \geq \frac{4}{a+b}.$$

Solution

Apply AM $\geq$ HM:

$$\frac{\frac{1}{a} + \frac{1}{b}}{2} \geq \frac{2}{a+b}.$$

Multiplying by 2,

$$\boxed{\frac{1}{a} + \frac{1}{b} \geq \frac{4}{a+b}.$$

Part B: Weighted Mean Inequality

Question 5

Find the greatest value of

$$x^2y^5$$

subject to

$$2x + 5y = 14.$$

Solution

Write

$$2x + 5y = x + x + y + y + y + y + y.$$

There are

$$7$$

terms.

Applying AM–GM,

$$\frac{14}{7} \geq \sqrt[7]{x^2 y^5}.$$

Hence,

$$2 \geq \sqrt[7]{x^2 y^5}.$$

Therefore,

$$\boxed{x^2 y^5 \leq 2^7 = 128.}$$

Equality when

$$x = y = 2.$$

Question 6

Find the greatest value of

$$x^4 y^3$$

subject to

$$4x + 3y = 21.$$

Solution

Write

$$4x + 3y = x + x + x + x + y + y + y.$$

Using AM–GM,

$$\frac{21}{7} \geq \sqrt[7]{x^4 y^3}.$$

Hence,

$$3 \geq \sqrt[7]{x^4 y^3}.$$

Therefore,

$$\boxed{\max(x^4 y^3) = 3^7.}$$

Equality when

$$x = y = 3.$$

Question 7

Show that

$$\left(\frac{a+b}{2a}\right)^a \left(\frac{a+b}{2b}\right)^b \leq 1.$$

Solution

Using the equivalent weighted AM–GM inequality,

$$\left(\frac{x_1}{\lambda_1}\right)^{\lambda_1} \left(\frac{x_2}{\lambda_2}\right)^{\lambda_2} \leq \left(\frac{x_1 + x_2}{\lambda_1 + \lambda_2}\right)^{\lambda_1 + \lambda_2},$$

choose

$$\begin{aligned} x_1 = x_2 &= \frac{a+b}{2}, \\ \lambda_1 = a, \lambda_2 &= b. \end{aligned}$$

Then

$$x_1 + x_2 = a + b = \lambda_1 + \lambda_2.$$

Hence,

$$\left(\frac{a+b}{2a}\right)^a \left(\frac{a+b}{2b}\right)^b \leq 1.$$

Question 8

Show that

$$\left(\frac{a+b+c}{3a}\right)^a \left(\frac{a+b+c}{3b}\right)^b \left(\frac{a+b+c}{3c}\right)^c \leq 1.$$

Solution

Take

$$x_1 = x_2 = x_3 = \frac{a+b+c}{3},$$

and

$$\lambda_1 = a, \lambda_2 = b, \lambda_3 = c.$$

Since

$$x_1 + x_2 + x_3 = a + b + c = \lambda_1 + \lambda_2 + \lambda_3,$$

Weighted AM–GM gives

$$\left(\frac{a+b+c}{3a}\right)^a \left(\frac{a+b+c}{3b}\right)^b \left(\frac{a+b+c}{3c}\right)^c \leq 1.$$

Question 9

Show that

$$\left(\frac{a+2b-c}{a}\right)^a \left(\frac{b+2c-a}{b}\right)^b \left(\frac{c+2a-b}{c}\right)^c \leq \left(\frac{2}{3}\right)^{-(a+b+c)}.$$

Solution

Take

$$\begin{aligned} x_1 &= a + 2b - c, \\ x_2 &= b + 2c - a, \\ x_3 &= c + 2a - b. \end{aligned}$$

Then

$$x_1 + x_2 + x_3 = 2(a + b + c).$$

Applying the equivalent weighted AM–GM,

$$\prod \left(\frac{x_i}{\lambda_i}\right)^{\lambda_i} \leq \left(\frac{2(a+b+c)}{a+b+c}\right)^{a+b+c},$$

giving

$$\left(\frac{a+2b-c}{a}\right)^a \left(\frac{b+2c-a}{b}\right)^b \left(\frac{c+2a-b}{c}\right)^c \leq 2^{a+b+c}.$$

Lecture 2: Summation of Algebraic Series

1. Sequence and Series

Definition (Sequence)

A **sequence** is an ordered collection of numbers.

Examples:

$$\begin{aligned} &2, 4, 6, 8, \dots \\ &1, 3, 9, 27, \dots \end{aligned}$$

Definition (Series)

A **series** is the sum of the terms of a sequence.

Examples

$$\begin{aligned} &2 + 4 + 6 + 8 + \dots \\ &1 + 3 + 9 + 27 + \dots \end{aligned}$$

2. Arithmetic Progression (A.P.)

Definition

A sequence is called an **Arithmetic Progression (A.P.)** if the difference between two consecutive terms is constant.

That constant is called the **common difference**.

Example

$$3, 7, 11, 15, \dots$$

Here,

$$d = 4.$$

nth Term

If

- first term = a ,
- common difference = d ,

then

$$T_n = a + (n - 1)d.$$

Proof

The sequence is

$$a, a + d, a + 2d, a + 3d, \dots$$

The second term is

$$a + d = a + (2 - 1)d.$$

The third term is

$$a + 2d = a + (3 - 1)d.$$

Similarly,
the n th term is

$$a + (n - 1)d.$$

Example 1

Find the 15th term of

$$5, 8, 11, \dots$$

Solution

Here,

$$a = 5, d = 3.$$

Hence

$$T_{15} = 5 + 14(3) = 47.$$

Sum of First n Terms of an A.P.

Suppose

$$a, a + d, a + 2d, \dots$$

Then

$$S_n = \frac{n}{2}[2a + (n - 1)d].$$

Equivalent form

$$S_n = \frac{n}{2}(a + l),$$

where

$$l = a + (n - 1)d$$

is the last (n th) term.

Proof

Let

$$S_n = a + (a + d) + \dots + [a + (n - 1)d] \dots \dots \dots (i)$$

Write the series in reverse order.

$$S_n = [a + (n - 1)d] + [a + (n - 2)d] + \dots + a \dots \dots \dots (ii)$$

Adding (i) and (ii),

$$2S_n = [2a + (n - 1)d] + [2a + (n - 1)d] + \dots$$

Or,

$$2S_n = n[2a + (n - 1)d]$$

Hence

$$S_n = \frac{n}{2}[2a + (n - 1)d].$$

Example 2

Find

$$5 + 8 + 11 + \dots + 62.$$

Solution

Here

$$a = 5, d = 3, l = 62.$$

Number of terms

$$62 = 5 + (n - 1)3$$

so

$$n = 20.$$

Therefore

$$S = \frac{20}{2} (5 + 62) = 670.$$

3. Geometric Progression (G.P.)

Definition

A sequence is called a **Geometric Progression** if the ratio of two consecutive terms is constant. That constant is called the **common ratio**.

Example

$$2, 6, 18, 54, \dots$$

Here

$$r = 3.$$

nth Term

$$T_n = ar^{n-1}.$$

Proof

Sequence

$$a, ar, ar^2, ar^3, \dots$$

The exponent increases by one each time.

Therefore

$$T_n = ar^{n-1}.$$

Example 3

Find the 8th term of

$$3, 6, 12, \dots$$

Solution

$$a = 3, r = 2.$$

Hence

$$T_8 = 3(2)^7 = 384.$$

Sum of First n Terms

For

$$r \neq 1,$$

$$S_n = a \frac{r^n - 1}{r - 1}.$$

Equivalent form

$$S_n = a \frac{1 - r^n}{1 - r}.$$

Proof

Let

$$S = a + ar + ar^2 + \dots + ar^{n-1}.$$

Multiply by r .

$$rS = ar + ar^2 + \dots + ar^n.$$

Subtract,

$$S - rS = a - ar^n.$$

Therefore,

$$S = a \frac{1 - r^n}{1 - r}.$$

Example 4

Find

$$2 + 6 + 18 + \dots + 1458.$$

Solution

Here

$$a = 2, r = 3.$$

Since

$$1458 = 2(3)^6,$$

there are

$$7$$

terms.

Hence

$$S = 2 \frac{3^7 - 1}{3 - 1} = 2186.$$

Infinite Geometric Series

If

$$|r| < 1,$$

then

$$S_{\infty} = \frac{a}{1-r}.$$

Proof

We know

$$S_n = a \frac{1-r^n}{1-r}.$$

Since

$$|r| < 1, \\ r^n \rightarrow 0$$

as

$$n \rightarrow \infty.$$

Hence

$$S_{\infty} = \frac{a}{1-r}.$$

Example 5

Find

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$$

Solution

$$a = 1, r = \frac{1}{2}.$$

Hence

$$S = \frac{1}{1 - \frac{1}{2}} = 2.$$

4. Special Algebraic Series

Series 1: Sum of Natural Numbers

Find

$$1 + 2 + 3 + \dots + n.$$

Formula

$$\frac{n(n+1)}{2}.$$

Proof

This is an A.P.

Here

$$a = 1, l = n.$$

Hence

$$S = \frac{n}{2}(1 + n).$$

Therefore

$$\frac{n(n+1)}{2}.$$

Or, you can do the proof by method of induction as well.

Mathematical Induction Principle

If a proposition is true for the number $n = 1$

and

If it can be shown that if the proposition is true for n ,
it will also be true for $n+1$

then

The proposition is true for all natural numbers: $1, 2, 3, \dots$

Example

Find

$$1 + 2 + \dots + 50.$$

Solution

$$S = \frac{50(51)}{2} = 1275.$$

Series 2: Sum of Squares

Find

$$1^2 + 2^2 + \dots + n^2.$$

Formula

$$\boxed{\frac{n(n+1)(2n+1)}{6}}.$$

Theorem 1

$$\boxed{1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}}.$$

Proof (By Mathematical Induction)

Let

$$P(n): 1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Step 1: Base Case

For $n = 1$,

LHS

$$= 1^2 = 1.$$

RHS

$$= \frac{1(2)(3)}{6} = 1.$$

Hence,

$$\text{LHS} = \text{RHS}.$$

Therefore, the result is true for $n = 1$.

Step 2: Induction Hypothesis

Assume the formula is true for $n = k$, i.e.,

$$1^2 + 2^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}.$$

Step 3: Induction Step

We must prove it for $n = k + 1$.

Consider

$$1^2 + 2^2 + \dots + k^2 + (k+1)^2.$$

Using the induction hypothesis,

$$= \frac{k(k+1)(2k+1)}{6} + (k+1)^2.$$

Take $(k+1)$ common:

$$= (k+1) \left(\frac{k(2k+1)}{6} + k+1 \right).$$

Write everything over a common denominator:

$$= (k+1) \left(\frac{k(2k+1) + 6(k+1)}{6} \right).$$

Simplify the numerator:

$$2k^2 + k + 6k + 6 = 2k^2 + 7k + 6.$$

Factorize:

$$2k^2 + 7k + 6 = (k+2)(2k+3).$$

Hence,

$$= \frac{(k+1)(k+2)(2k+3)}{6}.$$

Since

$$k+2 = (k+1) + 1,$$

and

$$2k+3 = 2(k+1) + 1,$$

we obtain

$$= \frac{(k+1)((k+1)+1)(2(k+1)+1)}{6}.$$

This is exactly the required formula for $n = k+1$.

Hence, by mathematical induction,

$$\boxed{1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}}.$$

Example

Find

$$1^2 + \dots + 10^2.$$

Answer

$$385.$$

Series 3: Sum of Cubes

Find

$$1^3 + 2^3 + \dots + n^3.$$

Formula

$$\left[\frac{n(n+1)}{2} \right]^2.$$

Proof (By Mathematical Induction)

Let

$$P(n): 1^3 + 2^3 + \dots + n^3 = \left(\frac{n(n+1)}{2} \right)^2.$$

Step 1: Base Case

For $n = 1$,

LHS

$$= 1.$$

RHS

$$= \left(\frac{1(2)}{2} \right)^2 = 1.$$

Thus,

$$\text{LHS} = \text{RHS}.$$

Step 2: Induction Hypothesis

Assume

$$1^3 + 2^3 + \dots + k^3 = \left(\frac{k(k+1)}{2} \right)^2.$$

Step 3: Induction Step

Consider

$$1^3 + \dots + k^3 + (k+1)^3.$$

Using the induction hypothesis,

$$= \left(\frac{k(k+1)}{2} \right)^2 + (k+1)^3.$$

Take $(k+1)^2$ common:

$$= (k+1)^2 \left(\frac{k^2}{4} + k + 1 \right).$$

Write over a common denominator:

$$= (k + 1)^2 \left(\frac{k^2 + 4k + 4}{4} \right).$$

Notice that

$$k^2 + 4k + 4 = (k + 2)^2.$$

Therefore,

$$= \frac{(k + 1)^2 (k + 2)^2}{4}.$$

This can be written as

$$= \left(\frac{(k + 1)(k + 2)}{2} \right)^2.$$

Since

$$k + 2 = (k + 1) + 1,$$

we have

$$= \left(\frac{(k + 1)((k + 1) + 1)}{2} \right)^2,$$

which is exactly the required formula for $n = k + 1$.

Hence, by mathematical induction,

$$1^3 + 2^3 + \dots + n^3 = \left(\frac{n(n + 1)}{2} \right)^2.$$

Example

Find

$$1^3 + \dots + 5^3.$$

Answer

$$225.$$

5. Recurring Number Series

Example 6

Find

$$3 + 33 + 333 + 3333.$$

Solution

Observe

$$\begin{aligned}
 3 &= 3 \\
 33 &= 3(10 + 1) \\
 333 &= 3(100 + 10 + 1)
 \end{aligned}$$

Hence

$$S = 3[(1) + (10 + 1) + (100 + 10 + 1) + (1000 + 100 + 10 + 1)].$$

Now combine the terms inside the brackets:

$$\begin{aligned}
 &1 + (10 + 1) + (100 + 10 + 1) + (1000 + 100 + 10 + 1) \\
 &= (1 + 1 + 1 + 1) + (10 + 10 + 10) + (100 + 100) + 1000 \\
 &= 4 + 30 + 200 + 1000 \\
 &= 1234.
 \end{aligned}$$

Simplify to obtain

$$S = 3(1234) = 3702.$$

Example 7

Find

$$5 + 55 + 555 + \dots$$

(first six terms).

Solution

Use

$$\begin{aligned}
 55 &= 5(11), \\
 555 &= 5(111),
 \end{aligned}$$

and express each term in powers of 10.

Exercise

- Find the sum of

$$7 + 10 + 13 + \dots + 76.$$

- Find the sum of

$$4 + 12 + 36 + \dots + 2916.$$

- Find

$$1 + 2 + \dots + 50.$$

- Find

$$1^2 + 2^2 + \dots + 20^2.$$

- Find

$$1^3 + 2^3 + \dots + 15^3.$$

6. Find

$$2 + 22 + 222 + 2222.$$

Lecture 3: Method of Difference and Finite Differences

1. Method of Difference (Telescoping Method)

Many algebraic series can be summed by expressing each term as the **difference of two simpler expressions**.

Most questions involve partial fractions.

Idea

Suppose

$$u_n = A_n - A_{n+1}.$$

Then

$$\sum_{n=1}^N u_n = (A_1 - A_2) + (A_2 - A_3) + \cdots + (A_N - A_{N+1}).$$

All middle terms cancel.

Hence,

$$\boxed{\sum_{n=1}^N (A_n - A_{n+1}) = A_1 - A_{N+1}.$$

This is called a **telescoping series**.

Example 1

Find

$$\sum_{r=1}^n \frac{1}{r(r+1)}.$$

Solution

Use partial fractions.

Let

$$\frac{1}{r(r+1)} = \frac{A}{r} + \frac{B}{r+1}.$$

Multiplying,

$$1 = A(r+1) + Br.$$

Comparing coefficients,

$$A = 1, B = -1.$$

Therefore,

$$\frac{1}{r(r+1)} = \frac{1}{r} - \frac{1}{r+1}.$$

Hence,

$$S = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right).$$

Everything cancels except

$$1 - \frac{1}{n+1}.$$

Therefore,

$$\boxed{S = \frac{n}{n+1}}.$$

Example 2

Find

$$\sum_{r=1}^n \frac{1}{(r+1)(r+2)}.$$

Solution

Resolve,

$$\frac{1}{(r+1)(r+2)} = \frac{1}{r+1} - \frac{1}{r+2}.$$

Hence,

$$S = \frac{1}{2} - \frac{1}{n+2}.$$

Therefore,

$$\boxed{S = \frac{n+1}{2(n+2)}}.$$

Example 3

Find

$$\sum_{r=1}^n \frac{2}{r(r+2)}.$$

Solution: Write

$$\frac{2}{r(r+2)} = \frac{1}{r} - \frac{1}{r+2}.$$

Write out the summation

$$\begin{aligned} S &= \sum_{r=1}^n \left(\frac{1}{r} - \frac{1}{r+2} \right) \\ &= \left(1 - \frac{1}{3} \right) + \left(\frac{1}{2} - \frac{1}{4} \right) + \left(\frac{1}{3} - \frac{1}{5} \right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+2} \right). \end{aligned}$$

Observe the cancellation

Now write the terms one below another:

$$\begin{aligned} S = & \quad 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n} \\ & - \frac{1}{3} - \frac{1}{4} - \frac{1}{5} - \cdots - \frac{1}{n+2}. \end{aligned}$$

Now cancel the common terms:

- $\frac{1}{3}$ cancels.
- $\frac{1}{4}$ cancels.
- $\frac{1}{5}$ cancels.
- ...
- $\frac{1}{n}$ cancels.

Only the first two positive terms and the last two negative terms remain.

Thus,

$$S = 1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2}.$$

Since

$$1 + \frac{1}{2} = \frac{3}{2},$$

we obtain

$$S = \frac{3}{2} - \frac{1}{n+1} - \frac{1}{n+2}.$$

Why do only these terms remain?

Here's the cancellation visually:

$$S = 1 + \frac{1}{2} + \frac{\cancel{1}}{\cancel{3}} + \frac{\cancel{1}}{\cancel{4}} + \frac{\cancel{1}}{\cancel{5}} + \cdots + \frac{\cancel{1}}{\cancel{n}} \\ - \frac{\cancel{1}}{\cancel{3}} - \frac{\cancel{1}}{\cancel{4}} - \frac{\cancel{1}}{\cancel{5}} - \cdots - \frac{\cancel{1}}{\cancel{n}} - \frac{1}{n+1} - \frac{1}{n+2}.$$

2. Finite Difference

Suppose

$$y_0, y_1, y_2, \dots$$

are values of a function at equally spaced arguments.

First Difference

The first forward difference is

$$\Delta y_i = y_{i+1} - y_i.$$

Example

Consider

$$2, 5, 10, 17, 26.$$

Then

y	2	5	10	17	26
Δy		3	5	7	9

Second Difference

$$\Delta^2 y_i = \Delta y_{i+1} - \Delta y_i.$$

For the above example,

$$\begin{aligned} 5 - 3 &= 2, \\ 7 - 5 &= 2, \\ 9 - 7 &= 2. \end{aligned}$$

Hence

y	2	5	10	17	26
Δy		3	5	7	9
$\Delta^2 y$			2	2	2

Observation

Constant second difference

$\Rightarrow$

quadratic polynomial.

Example

Construct the difference table for

1, 8, 27, 64, 125.

Solution

y	1	8	27	64	125
Δy		7	19	37	61
$\Delta^2 y$			12	18	24
$\Delta^3 y$				6	6

Therefore,
the third difference is constant.

Hence,
the sequence comes from a cubic polynomial.

3. Differences of Polynomials

Theorem

If

$$f(x)$$

is a polynomial of degree

$$n,$$

then

$$\Delta^n f(x) = \text{constant},$$

and

$$\Delta^{n+1} f(x) = 0.$$

Illustration

Polynomial	Constant Difference
Degree 1	First
Degree 2	Second
Degree 3	Third
Degree 4	Fourth

Example

For

$$f(x) = x^2,$$

values are

$$0, 1, 4, 9, 16.$$

Difference table

y	0	1	4	9	16
Δ		1	3	5	7
Δ^2			2	2	2

Hence,

second difference is constant.

4. Finite Difference Equation

A finite difference equation relates successive terms of a sequence.

Example,

$$a_{n+1} - a_n = 5.$$

Example 1

Solve

$$a_{n+1} - a_n = 4,$$

given

$$a_1 = 3.$$

Solution

Successive terms are

$$3, 7, 11, 15, \dots$$

Hence,

$$a_n = 3 + 4(n - 1).$$

Example 2

Solve

$$a_{n+1} - a_n = 2n,$$

with

$$a_1 = 1.$$

Solution

Successively,

$$a_2 = 3,$$

$$\begin{aligned}a_3 &= 7, \\a_4 &= 13.\end{aligned}$$

Recognize

$$a_n = n^2 - n + 1.$$

Starting with

$$a_{n+1} - a_n = 2n,$$

write the first few equations:

$$\begin{aligned}a_2 - a_1 &= 2, \\a_3 - a_2 &= 4, \\a_4 - a_3 &= 6, \\\vdots \\a_n - a_{n-1} &= 2(n-1).\end{aligned}$$

Now add all these equations:

$$(a_2 - a_1) + (a_3 - a_2) + \cdots + (a_n - a_{n-1}) = 2(1 + 2 + \cdots + n - 1).$$

The left-hand side telescopes:

$$a_n - a_1 = 2 \cdot \frac{(n-1)n}{2} = n(n-1).$$

Since $a_1 = 1$,

$$a_n - 1 = n(n-1),$$

so

$$\boxed{a_n = n^2 - n + 1.}$$

Verification:

$$(n+1)^2 - (n+1) + 1 - (n^2 - n + 1) = 2n.$$

Hence,

$$\boxed{a_n = n^2 - n + 1.}$$

Exercise

1. Find

$$\sum_{r=1}^n \frac{1}{r(r+2)}.$$

2. Find

$$\sum_{r=1}^n \frac{3}{(r+1)(r+4)}.$$

3. Construct the difference table for

1, 4, 9, 16, 25.

4. Construct the difference table for

2, 9, 28, 65, 126.

5. Solve

$$a_{n+1} - a_n = 6,$$

given

$$a_1 = 5.$$

6. Solve

$$a_{n+1} - a_n = 2n + 1,$$

given

$$a_1 = 2.$$

Lecture 4: Differences of Polynomials

1. Forward Difference Operator

Suppose

$$f(x)$$

is a function defined at equally spaced values of x .

The **forward difference operator** is defined by

$$\Delta f(x) = f(x+h) - f(x),$$

where

$$h$$

is the interval of difference.

Usually,

$$h = 1.$$

Example

If

$$f(x) = x^2,$$

then

$$\Delta f(x) = (x + 1)^2 - x^2.$$

Expanding,

$$= x^2 + 2x + 1 - x^2 = 2x + 1.$$

Hence

$$\boxed{\Delta(x^2) = 2x + 1.}$$

2. Difference of a Constant

Let

$$f(x) = c.$$

Then

$$\Delta c = c - c = 0.$$

Hence

$$\boxed{\Delta(c) = 0.}$$

3. Difference of a Linear Polynomial

Let

$$f(x) = ax + b.$$

Then

$$\begin{aligned}\Delta(ax + b) &= [a(x + 1) + b] - (ax + b) \\ &= a.\end{aligned}$$

Therefore,

$$\boxed{\Delta(ax + b) = a.}$$

Example

Find

$$\Delta(5x + 3).$$

Solution

$$5(x + 1) + 3 - (5x + 3) = 5.$$

Hence

5.

4. Difference of a Quadratic

Let

$$f(x) = x^2.$$

Already,

$$\Delta(x^2) = 2x + 1.$$

Again,

$$\Delta^2(x^2) = \Delta(2x + 1).$$

Now,

$$2(x + 1) + 1 - (2x + 1) = 2.$$

Hence

$$\Delta^2(x^2) = 2.$$

Also,

$$\Delta^3(x^2) = 0.$$

Example

Find

$$\Delta(x^2 + 3x + 5).$$

Solution

$$(x + 1)^2 + 3(x + 1) + 5 - (x^2 + 3x + 5).$$

Simplifying,

$$2x + 4.$$

Hence

$$\Delta(x^2 + 3x + 5) = 2x + 4.$$

5. Difference of a Cubic

Let

$$f(x) = x^3.$$

Then

$$\begin{aligned}\Delta(x^3) &= (x+1)^3 - x^3 \\ &= 3x^2 + 3x + 1.\end{aligned}$$

Hence

$$\boxed{\Delta(x^3) = 3x^2 + 3x + 1.}$$

Again,

$$\Delta^2(x^3) = 6x + 6.$$

Again,

$$\Delta^3(x^3) = 6.$$

Finally,

$$\boxed{\Delta^4(x^3) = 0.}$$

Example

Find

$$\Delta(x^3 + 2x^2 - 4).$$

Solution

$$(x+1)^3 + 2(x+1)^2 - 4 - (x^3 + 2x^2 - 4).$$

Simplifying,

$$3x^2 + 7x + 3.$$

Hence

$$\boxed{\Delta(x^3 + 2x^2 - 4) = 3x^2 + 7x + 3.}$$

6. General Result

Theorem

If

$$f(x)$$

is a polynomial of degree

$$n,$$

then

$$\boxed{\Delta f(x)}$$

is a polynomial of degree

$$n - 1.$$

Proof

Suppose

$$f(x) = a_n x^n + \dots .$$

Then

$$\Delta f(x) = a_n [(x + 1)^n - x^n] + \dots .$$

Using the Binomial Theorem,

$$(x + 1)^n = x^n + nx^{n-1} + \dots .$$

Hence

$$(x + 1)^n - x^n = nx^{n-1} + \dots .$$

Therefore

$$\boxed{\Delta f(x)}$$

has degree

$$n - 1.$$

Corollary

Applying the operator repeatedly,

$$\boxed{\Delta^n f(x) = \text{constant.}}$$

and

$$\boxed{\Delta^{n+1} f(x) = 0.}$$

Important Table

Polynomial	First Difference	Highest Constant Difference
c	0	First
$ax + b$	Constant	First
$ax^2 + bx + c$	Linear	Second
$ax^3 + bx^2 + cx + d$	Quadratic	Third
Degree n	Degree $n - 1$	n th

Example 1

Determine the degree of the polynomial whose difference table is

Values 2 9 28 65 126

Δ 7 19 37 61

Δ^2 12 18 24

Δ^3 6 6

Solution

Third difference is constant.

Therefore

$$\boxed{\text{Degree} = 3.}$$

Example 2

Evaluate

$$\Delta(x^4).$$

Solution

Using the Binomial Theorem,

$$(x + 1)^4 = x^4 + 4x^3 + 6x^2 + 4x + 1.$$

Hence

$$\boxed{\Delta(x^4) = 4x^3 + 6x^2 + 4x + 1.}$$

Example 3

Show that

$$\Delta(x^n)$$

is a polynomial of degree

$$n - 1.$$

Solution

Using the Binomial Theorem,

$$(x + 1)^n = x^n + nx^{n-1} + \dots .$$

Therefore,

$$(x + 1)^n - x^n = nx^{n-1} + \dots ,$$

whose leading term is

$$nx^{n-1}.$$

Hence,

$$\boxed{\Delta(x^n)}$$

has degree

$$n - 1.$$

Exercises

1. Find

$$\Delta(x^2 - 5x + 2).$$

2. Find

$$\Delta(x^3 - 2x + 7).$$

3. Find

$$\Delta(x^4 + x^2).$$

4. Show that

$$\Delta(x^5) = 5x^4 + 10x^3 + 10x^2 + 5x + 1.$$

5. Prove that the fourth difference of every quartic polynomial is constant.

6. Determine the degree of a polynomial whose fifth difference is constant.

Lecture 5: Polynomial Equations

1. Polynomial

Definition

A polynomial in one variable is an expression of the form

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where

- a_i are constants,
- $a_n \neq 0$,
- n is a non-negative integer.

Examples

Polynomial

$$3x^4 - 5x^2 + 2x - 7$$

Polynomial

$$x^5 + 4x^3 - 2$$

Not Polynomial

$$x^{-2} + 3$$

(negative exponent)

Not Polynomial

$$\sqrt{x} + 1$$

(fractional exponent)

Not Polynomial

$$\frac{1}{x+1}$$

(variable in denominator)

Degree of a Polynomial

The degree is the highest exponent.

Examples

Polynomial	Degree
$5x^3 - 2x + 1$	3
$7x^5 + 2$	5
4	0
0	Undefined

Example 1

Find the degree of

$$5x^7 - 2x^4 + 8x - 9.$$

Solution

Highest exponent is

7.

Hence

$$\deg(P) = 7.$$

2. Polynomial Equation

Definition

A polynomial equation is an equation of the form

$$P(x) = 0.$$

Examples

$$\begin{aligned}x^2 - 5x + 6 &= 0 \\x^3 - 4x + 1 &= 0.\end{aligned}$$

Root (Zero) of a Polynomial

A number

$$\alpha$$

is called a root if

$$P(\alpha) = 0.$$

Example

For

$$\begin{aligned}P(x) &= x^2 - 5x + 6, \\P(2) &= 0, \\P(3) &= 0.\end{aligned}$$

Hence

$$\boxed{2,3}$$

are roots.

Example 2

Determine whether

$$2$$

is a root of

$$P(x) = x^3 - 3x + 2.$$

Solution

$$P(2) = 8 - 6 + 2 = 4.$$

Since

$$\begin{aligned}P(2) &\neq 0, \\ \boxed{2 \text{ is not a root.}}\end{aligned}$$

3. Factor Theorem**Theorem**

If

$$P(a) = 0,$$

then

$$\boxed{x - a}$$

is a factor of

$$P(x).$$

Conversely,
if

$$x - a$$

is a factor,
then

$$P(a) = 0.$$

Proof

Suppose

$$x - a$$

is a factor.
Then

$$P(x) = (x - a)Q(x).$$

Putting

$$x = a,$$

gives

$$P(a) = 0.$$

Conversely,
by the Division Algorithm,

$$P(x) = (x - a)Q(x) + R,$$

where

$$R$$

is constant.
Putting

$$\begin{aligned} x &= a, \\ P(a) &= R. \end{aligned}$$

Hence,
if

$$P(a) = 0,$$

then

$$R = 0,$$

so

$$x - a$$

is a factor.

Example 3

Show that

$$x - 2$$

is a factor of

$$x^3 - 4x^2 + x + 6.$$

Solution

$$P(2) = 8 - 16 + 2 + 6 = 0.$$

Hence,

$$\boxed{x - 2 \text{ is a factor.}}$$

Example 4

Determine whether

$$x + 1$$

is a factor of

$$2x^3 + x^2 - 5x + 2.$$

Solution

Substitute

$$x = -1.$$

$$P(-1) = -2 + 1 + 5 + 2 = 6.$$

Therefore,

$$\boxed{x + 1 \text{ is not a factor.}}$$

4. Remainder Theorem**Theorem**

When

$$P(x)$$

is divided by

$$x - a,$$

the remainder is

$$\boxed{P(a)}.$$

Proof

Using the Division Algorithm,

$$P(x) = (x - a)Q(x) + R.$$

Putting

$$\begin{aligned}x &= a, \\ P(a) &= R.\end{aligned}$$

Hence,

$$\boxed{\text{Remainder} = P(a)}.$$

Example 5

Find the remainder when

$$2x^4 - 3x + 5$$

is divided by

$$x - 2.$$

Solution

Compute

$$P(2) = 2(16) - 6 + 5 = 31.$$

Hence

$$\boxed{\text{Remainder} = 31}.$$

Example 6

Find the remainder when

$$x^5 + 2x^3 - 4$$

is divided by

$$x + 2.$$

Solution

Use

$$\begin{aligned}a &= -2. \\ P(-2) &= -32 - 16 - 4 = -52.\end{aligned}$$

Hence

-52.

5. Synthetic Division (Horner's Method)

Synthetic division provides a quick way to divide a polynomial by a linear factor.

Example

Divide

$$x^3 - 6x^2 + 11x - 6$$

by

$$x - 1.$$

Solution

Step 1. Write the coefficients

The coefficients of the polynomial are

$$1, -6, 11, -6.$$

Since the divisor is

$$x - 1,$$

solve

$$x - 1 = 0,$$

which gives

$$x = 1.$$

Therefore, we use **1** in the synthetic division.

Step 2. Draw the synthetic division table

1	1	-6	11	-6

Step 3. Bring down the first coefficient

The first coefficient is always copied directly.

1	1	-6	11	-6
	1			

Step 4. Multiply and add

Multiply the number just written by 1.

$$1 \times 1 = 1.$$

Write it under the next coefficient.

1	1	-6	11	-6
		1		
	1			

Now add vertically.

$$-6 + 1 = -5.$$

1	1	-6	11	-6
		1		
	1	-5		

Step 5. Repeat

Multiply

$$-5 \times 1 = -5.$$

Place it below 11.

1	1	-6	11	-6
		1	-5	
	1	-5		

Add:

$$11 + (-5) = 6.$$

1	1	-6	11	-6
		1	-5	
	1	-5	6	

Step 6. Repeat once more

Multiply

$$6 \times 1 = 6.$$

Write it below -6.

1	1	-6	11	-6
		1	-5	6
	1	-5	6	

Add:

$$-6 + 6 = 0.$$

Therefore,

$$\begin{array}{r|rrrrr} 1 & 1 & -6 & 11 & -6 & \\ \hline & & 1 & -5 & 6 & \\ \hline & 1 & -5 & 6 & 0 & \end{array}$$

Step 7. Interpret the result

Since the original polynomial is of degree 3, the quotient must be of degree 2.

The numbers

$$1, -5, 6$$

are the coefficients of the quotient.

Hence,

$$\boxed{x^2 - 5x + 6.}$$

The last entry,

$$\boxed{0},$$

is the **remainder**.

Therefore,

$$x^3 - 6x^2 + 11x - 6 = (x - 1)(x^2 - 5x + 6).$$

Example 8

Divide

$$2x^3 + 3x^2 - 5x + 4$$

by

$$x + 2.$$

Solution

Use

$$-2.$$

$$\begin{array}{r|rrrrr} -2 & 2 & 3 & -5 & 4 & \\ \hline & & -4 & 2 & 6 & \\ \hline & 2 & -1 & -3 & 10 & \end{array}$$

Therefore,
Quotient

$$2x^2 - x - 3$$

Remainder

$$10.$$

6. Fundamental Theorem of Algebra

Theorem

Every polynomial of degree

$$n$$

has exactly

$$n$$

roots in the complex number system, counting multiplicities.

Examples

Example 1

$$x^2 + 1 = 0.$$

Roots

$$\pm i.$$

Example 2

$$x^3 - 1 = 0.$$

There are exactly

$$3$$

roots.

Example 3

$$(x - 2)^3(x + 1)^2 = 0.$$

Degree

$$5.$$

Roots

$$2, 2, 2, -1, -1.$$

Exactly five roots counting multiplicity.

Exercises

1. Find the degree of $5x^8 - 2x^3 + 7$.
2. Determine whether 3 is a root of $x^3 - 10x + 3$.
3. Show that $x - 1$ is a factor of $x^3 - 4x^2 + x + 2$.
4. Find the remainder when $3x^5 + 2x - 7$ is divided by $x - 2$.
5. Divide $x^3 - 7x + 6$ by $x - 1$ using synthetic division.
6. Divide $2x^4 - x^2 + 5$ by $x + 1$.

Lecture 6: Relations Between Roots and Coefficients (Vieta's Formula)

1. Introduction

Consider the quadratic equation

$$ax^2 + bx + c = 0, a \neq 0.$$

Suppose its roots are

$$\alpha, \beta.$$

Then

$$a(x - \alpha)(x - \beta) = ax^2 - a(\alpha + \beta)x + a\alpha\beta.$$

Comparing coefficients,
we obtain the relations between the roots and coefficients.

2. Vieta's Formula for Quadratic Equation

If

$$ax^2 + bx + c = 0$$

has roots

$$\alpha, \beta,$$

then

$$\boxed{\alpha + \beta = -\frac{b}{a},}$$

and

$$\boxed{\alpha\beta = \frac{c}{a}.}$$

Proof

We have

$$ax^2 + bx + c = a(x - \alpha)(x - \beta).$$

Expanding,

$$= ax^2 - a(\alpha + \beta)x + a\alpha\beta.$$

Comparing coefficients,
Coefficient of x

$$-a(\alpha + \beta) = b,$$

therefore

$$\boxed{\alpha + \beta = -\frac{b}{a}.$$

Constant term

$$a\alpha\beta = c.$$

Hence

$$\boxed{\alpha\beta = \frac{c}{a}.$$

Example 1

Find

$$\alpha + \beta, \alpha\beta$$

for

$$3x^2 - 5x + 7 = 0.$$

Solution

$$a = 3, b = -5, c = 7.$$

Therefore

$$\boxed{\alpha + \beta = \frac{5}{3},$$

and

$$\boxed{\alpha\beta = \frac{7}{3}.$$

3. Cubic Equation

Suppose

$$ax^3 + bx^2 + cx + d = 0$$

has roots

$$\alpha, \beta, \gamma.$$

Then

$$a(x - \alpha)(x - \beta)(x - \gamma) = ax^3 - a(\alpha + \beta + \gamma)x^2 + a(\alpha\beta + \beta\gamma + \gamma\alpha)x - a\alpha\beta\gamma.$$

Comparing coefficients,

Vieta's Formula

$$\alpha + \beta + \gamma = -\frac{b}{a}.$$

$$\sum \alpha\beta = \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}.$$

$$\alpha\beta\gamma = -\frac{d}{a}.$$

Example 3

For

$$x^3 - 4x^2 + 5x - 2 = 0,$$

find

$$\begin{aligned}\sum \alpha &= \alpha + \beta + \gamma, \\ \sum \alpha\beta &= \alpha\beta + \beta\gamma + \gamma\alpha,\end{aligned}$$

and

$$\alpha\beta\gamma.$$

Solution

$$a = 1, b = -4, c = 5, d = -2.$$

Hence

$$\alpha + \beta + \gamma = 4,$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = 5,$$

$$\alpha\beta\gamma = 2.$$

Example: If a, b, c are the roots of $x^3 + px^2 + qx + r = 0$, find $S = \frac{b^2+c^2}{bc} + \frac{c^2+a^2}{ca} + \frac{a^2+b^2}{ab}$.

Solution

We have $\frac{b^2+c^2}{bc} = \frac{b}{c} + \frac{c}{b}$, therefore $S = \frac{a}{b} + \frac{b}{a} + \frac{b}{c} + \frac{c}{b} + \frac{c}{a} + \frac{a}{c}$.

Taking the LCM abc , $S = \frac{a^2c+b^2c+b^2a+c^2a+c^2b+a^2b}{abc}$.

Now use the standard identity

$$a^2b + a^2c + b^2a + b^2c + c^2a + c^2b = (a+b+c)(ab+bc+ca) - 3abc.$$

$$\text{Hence, } S = \frac{(a+b+c)(ab+bc+ca) - 3abc}{abc}.$$

From Vieta's formulas, $a+b+c = -p$, $ab+bc+ca = q$, $abc = -r$.

Therefore,

$$\begin{aligned} S &= \frac{(-p)q - 3(-r)}{-r} \\ &= \frac{-pq + 3r}{-r} \\ &= \frac{pq - 3r}{r}. \end{aligned}$$

$$\text{Hence, } \boxed{\frac{b^2+c^2}{bc} + \frac{c^2+a^2}{ca} + \frac{a^2+b^2}{ab} = \frac{pq-3r}{r}}.$$

Example: If the roots of the equation $x^3 + 3x^2 + 4x + 5 = 0$ are a, b, c , find the equation whose roots are $a+b, b+c, c+a$.

Solution:

Let, $\alpha = a+b, \beta = b+c, \gamma = c+a$. We first find the symmetric sums of α, β, γ .

For $x^3 + 3x^2 + 4x + 5 = 0$, we have, $a+b+c = -3$, $ab+bc+ca = 4$, $abc = -5$.

Sum of the new roots:

$$\begin{aligned} \alpha + \beta + \gamma &= (a+b) + (b+c) + (c+a) \\ &= 2(a+b+c) \\ &= 2(-3) = -6. \end{aligned}$$

Sum of the products of the new roots:

$$\alpha\beta + \beta\gamma + \gamma\alpha = (a+b)(b+c) + (b+c)(c+a) + (c+a)(a+b).$$

Using the identity

$$(a+b)(b+c) + (b+c)(c+a) + (c+a)(a+b) = (a+b+c)^2 + ab + bc + ca,$$

we get

$$= (-3)^2 + 4 = 9 + 4 = 13.$$

Product of the new roots: $\alpha\beta\gamma = (a+b)(b+c)(c+a).$

Use the standard identity

$$(a+b)(b+c)(c+a) = (a+b+c)(ab+bc+ca) - abc.$$

Hence,

$$= (-3)(4) - (-5) = -12 + 5 = -7.$$

The required equation is

$$x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \beta\gamma + \gamma\alpha)x - \alpha\beta\gamma = 0.$$

Substituting the values, $x^3 - (-6)x^2 + 13x - (-7) = 0.$

Therefore,

$$\boxed{x^3 + 6x^2 + 13x + 7 = 0.}$$

Problem

If α, β, γ are the roots of

$$x^3 + ax^2 + bx + c = 0,$$

find the equation whose roots are

$$\frac{1}{\beta\gamma}, \frac{1}{\gamma\alpha}, \frac{1}{\alpha\beta}.$$

Solution

Let the new roots be

$$u = \frac{1}{\beta\gamma}, v = \frac{1}{\gamma\alpha}, w = \frac{1}{\alpha\beta}.$$

We find their symmetric sums.

Step 1: Apply Vieta's formulas

For

$$\begin{aligned}x^3 + ax^2 + bx + c &= 0, \\ \alpha + \beta + \gamma &= -a, \\ \alpha\beta + \beta\gamma + \gamma\alpha &= b, \\ \alpha\beta\gamma &= -c.\end{aligned}$$

Step 2: Sum of the new roots

$$\begin{aligned}
 u + v + w &= \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha} + \frac{1}{\alpha\beta} \\
 &= \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma} \\
 &= \frac{-a}{-c} \\
 &= \frac{a}{c}.
 \end{aligned}$$

Step 3: Sum of the products of the new roots

$$\begin{aligned}
 uv + vw + wu &= \frac{1}{\alpha\beta^2\gamma} + \frac{1}{\alpha\beta\gamma^2} + \frac{1}{\alpha^2\beta\gamma} \\
 &= \frac{\alpha + \beta + \gamma}{(\alpha\beta\gamma)^2} \\
 &= \frac{-a}{c^2}.
 \end{aligned}$$

Since

$$\alpha\beta + \beta\gamma + \gamma\alpha = b,$$

it is easier to write

$$\begin{aligned}
 uv + vw + wu &= \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{(\alpha\beta\gamma)^2} \\
 &= \frac{b}{c^2}.
 \end{aligned}$$

Step 4: Product of the new roots

$$uvw = \frac{1}{(\alpha\beta\gamma)^2} = \frac{1}{c^2}.$$

Step 5: Form the required equation

The required equation is

$$x^3 - (u + v + w)x^2 + (uv + vw + wu)x - uvw = 0.$$

Substituting,

$$x^3 - \frac{a}{c}x^2 + \frac{b}{c^2}x - \frac{1}{c^2} = 0.$$

Multiplying throughout by c^2 ,

$$\boxed{c^2x^3 - acx^2 + bx - 1 = 0.}$$

Identities Used

1. $(a + b)(b + c) + (b + c)(c + a) + (c + a)(a + b) = (a + b + c)^2 + ab + bc + ca.$

2. $(a + b)(b + c)(c + a) = (a + b + c)(ab + bc + ca) - abc.$

(Memorize these formulas because they are frequently used.)

4. Quartic Equation

For

$$ax^4 + bx^3 + cx^2 + dx + e = 0,$$

the relations are

$$\alpha + \beta + \gamma + \delta = -\frac{b}{a}.$$

$$\sum \alpha\beta = \frac{c}{a}.$$

$$\sum \alpha\beta\gamma = -\frac{d}{a}.$$

$$\alpha\beta\gamma\delta = \frac{e}{a}.$$

6. Definition

A function of the roots is called a **symmetric function** if its value remains unchanged under every permutation (interchange) of the roots.

In other words, changing the order of the roots does not change the value of the function.

Example 1

Consider

$$f(\alpha, \beta, \gamma) = \alpha + \beta + \gamma.$$

Interchange α and β :

$$f(\beta, \alpha, \gamma) = \beta + \alpha + \gamma = \alpha + \beta + \gamma.$$

Hence,

$$\alpha + \beta + \gamma$$

is a symmetric function.

Example 2

Consider

$$\alpha\beta + \beta\gamma + \gamma\alpha.$$

Exchange α and γ .

Then

$$\gamma\beta + \beta\alpha + \alpha\gamma = \alpha\beta + \beta\gamma + \gamma\alpha.$$

The value is unchanged.

Therefore,

$$\boxed{\alpha\beta + \beta\gamma + \gamma\alpha}$$

is also symmetric.

Example 3

Consider

$$\alpha^2 + \beta^2 + \gamma^2.$$

Changing the order of the roots merely changes the order of the terms.

For example,

$$\beta^2 + \gamma^2 + \alpha^2 = \alpha^2 + \beta^2 + \gamma^2.$$

Hence,

$$\boxed{\alpha^2 + \beta^2 + \gamma^2}$$

is symmetric.

Example 4 (Not Symmetric)

Consider

$$\alpha - \beta.$$

Interchange α and β :

$$\beta - \alpha.$$

Clearly,

$$\beta - \alpha \neq \alpha - \beta$$

unless $\alpha = \beta$.

Therefore,

$$\boxed{\alpha - \beta}$$

is **not** a symmetric function.

Example 5 (Not Symmetric)

Consider

$$\frac{\alpha}{\beta}.$$

Interchanging α and β gives

$$\frac{\beta}{\alpha},$$

which is generally different.

Hence,

$$\boxed{\frac{\alpha}{\beta}}$$

is **not** symmetric.

Why Are Symmetric Functions Important?

Suppose the polynomial is

$$x^3 - 6x^2 + 11x - 6 = 0,$$

whose roots are

$$\alpha, \beta, \gamma.$$

Without solving the equation, Vieta's formulas give

$$\begin{aligned}\alpha + \beta + \gamma &= 6, \\ \alpha\beta + \beta\gamma + \gamma\alpha &= 11, \\ \alpha\beta\gamma &= 6.\end{aligned}$$

Now suppose we want to find

$$\alpha^2 + \beta^2 + \gamma^2.$$

Using the identity

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha),$$

we obtain

$$= 6^2 - 2(11) = 36 - 22 = 14.$$

Thus,

$$\boxed{\alpha^2 + \beta^2 + \gamma^2 = 14.}$$

Notice that we **never solved the cubic equation**. We used only the symmetric functions of the roots and Vieta's formulas.

Example 5

If

$$\alpha, \beta$$

are roots of

$$x^2 - 7x + 10 = 0,$$

find

$$\alpha^2 + \beta^2.$$

Solution

Use

$$(\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta.$$

Hence

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta.$$

Now

$$\begin{aligned}\alpha + \beta &= 7, \\ \alpha\beta &= 10.\end{aligned}$$

Therefore

$$49 - 20 = 29.$$

Hence

$$\boxed{29.}$$

Example 6

Find

$$\frac{1}{\alpha} + \frac{1}{\beta}.$$

Solution

$$= \frac{\alpha + \beta}{\alpha\beta}.$$

Hence

$$\boxed{\frac{-b/a}{c/a} = -\frac{b}{c}.$$

Example 7

If

$$\alpha, \beta$$

are roots of

$$2x^2 - 3x + 5 = 0,$$

find

$$\alpha^2\beta + \alpha\beta^2.$$

Solution

Factor

$$= \alpha\beta(\alpha + \beta).$$

Now

$$\alpha\beta = \frac{5}{2},$$
$$\alpha + \beta = \frac{3}{2}.$$

Hence

$$\boxed{\frac{15}{4}}.$$

7. Roots in Arithmetic Progression

Suppose the roots are

$$a - d, a, a + d.$$

Then

$$\text{Sum} = 3a.$$

Example: If $x^3 - 12x^2 + 44x - k = 0$ has roots in A.P., find k .

Solution

By Vieta's formula, $\alpha + \beta + \gamma = 12$. Since the roots are $a - d, a, a + d$, their sum is $(a - d) + a + (a + d) = 3a$.

The $-d$ and $+d$ cancel, so

$$3a = 12.$$

Hence,

$$a = \frac{12}{3} = 4.$$

Hence, the roots are

$$4 - d, 4, 4 + d.$$

Product

$$= 4(16 - d^2).$$

Using

$$(4 - d)4 + 4(4 + d) + (4 + d)(4 - d) = 44 = \alpha\beta + \beta\gamma + \gamma\alpha,$$

gives

$$48 - d^2 = 44.$$

Hence

$$d^2 = 4.$$

Therefore

$$\alpha\beta\gamma = 4(16 - d^2) = 4(12) = 48.$$

Hence

$$\boxed{k = 48.}$$

Example: Prove that if the roots of $x^3 - px^2 + qx - r = 0$ are in Arithmetic Progression (A.P.), then $\boxed{2p^3 + 27r - 9pq = 0.}$

Solution:

Let the three roots be $a - d, a, a + d$, since they are in A.P.

For the equation $x^3 - px^2 + qx - r = 0$, Vieta gives

$$\begin{aligned}(a - d) + a + (a + d) &= p, \\ (a - d)a + a(a + d) + (a - d)(a + d) &= q, \\ (a - d)a(a + d) &= r.\end{aligned}$$

Sum of roots: $3a = p$, so $\boxed{a = \frac{p}{3}.}$

Sum of products of two roots:

$$\begin{aligned}q &= a(a - d) + a(a + d) + (a - d)(a + d) \\ &= (a^2 - ad) + (a^2 + ad) + (a^2 - d^2) \\ &= 3a^2 - d^2.\end{aligned}$$

Hence $\boxed{d^2 = 3a^2 - q.}$ Since $a = \frac{p}{3}$, we get $\boxed{d^2 = \frac{p^2}{3} - q.}$

Product of roots:

$$\begin{aligned}r &= (a - d)a(a + d) \\ &= a(a^2 - d^2).\end{aligned}$$

Substitute $d^2 = 3a^2 - q$, to obtain

$$\begin{aligned} r &= a[a^2 - (3a^2 - q)] \\ &= a(q - 2a^2). \end{aligned}$$

Using $a = \frac{p}{3}$, we have $r = \frac{p}{3}\left(q - \frac{2p^2}{9}\right)$.

Therefore,

$$r = \frac{pq}{3} - \frac{2p^3}{27}.$$

Multiplying both sides by 27: $27r = 9pq - 2p^3$.

Hence,

$$2p^3 + 27r - 9pq = 0.$$

Roots in Geometric Progression (G.P.)

Concept

If three roots are in geometric progression, we write them as

$$\frac{a}{r}, a, ar,$$

where

- a is the middle term (geometric mean),
- r is the common ratio ($r \neq 0$).

This is because

$$\frac{a}{r}, a, ar$$

satisfy

$$\frac{a}{a/r} = \frac{ar}{a} = r.$$

Hence, the ratio of consecutive terms is constant.

Example

Suppose the roots of

$$x^3 - 14x^2 + 56x - k = 0$$

are in G.P.

Let the roots be

$$\frac{a}{r}, a, ar.$$

Using Vieta,

$$\frac{a}{r} + a + ar = 14.$$

Also,

$$\frac{a}{r} \times a \times ar = a^3 = k.$$

The remaining Vieta relation,

$$\frac{a^2}{r} + a^2r + a^2 = 56,$$

allows us to determine a , r , and k .

Observation

Unlike A.P., the middle root **cannot** usually be found immediately from the sum of the roots because the unknown ratio r is also involved.

Roots in Harmonic Progression (H.P.)

Concept

Three numbers are in harmonic progression if their reciprocals are in arithmetic progression. Therefore, if the roots are in H.P., then

$$\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$$

are in A.P.

Hence, we let

$$\frac{1}{\alpha} = a - d, \frac{1}{\beta} = a, \frac{1}{\gamma} = a + d.$$

Taking reciprocals,

$$\boxed{\alpha = \frac{1}{a - d}, \beta = \frac{1}{a}, \gamma = \frac{1}{a + d}.$$

This is the standard representation for roots in H.P.

Example

Suppose the roots of

$$x^3 - 6x^2 + 11x - k = 0$$

are in H.P.

Write

$$\frac{1}{\alpha} = a - d, \frac{1}{\beta} = a, \frac{1}{\gamma} = a + d.$$

Then,

$$\alpha = \frac{1}{a - d}, \beta = \frac{1}{a}, \gamma = \frac{1}{a + d}.$$

Now apply Vieta's formulas:

$$\begin{aligned}\alpha + \beta + \gamma &= 6, \\ \alpha\beta + \beta\gamma + \gamma\alpha &= 11, \\ \alpha\beta\gamma &= k.\end{aligned}$$

These equations determine a , d , and k .

Summary

Progression

Assume the roots are

Arithmetic Progression (A.P.) $a - d, a, a + d$

Geometric Progression (G.P.) $\frac{a}{r}, a, ar$

Harmonic Progression (H.P.) $\frac{1}{a-d}, \frac{1}{a}, \frac{1}{a+d}$

Exercise

1. Find the sum and product of the roots of $5x^2 - 8x + 3 = 0$.
2. Construct the equation whose roots are 4, -2.
3. Find $\alpha^2 + \beta^2$ if $3x^2 - 5x + 2 = 0$.
4. Find $\frac{1}{\alpha} + \frac{1}{\beta}$.
5. Find $\alpha^2\beta + \alpha\beta^2$ if $x^2 - 6x + 5 = 0$.

Lecture 7: Transformation of Equations and Symmetric Functions of Roots

1. Transformation of Equations

Definition

Suppose

$$f(x) = 0$$

has roots

$$\alpha, \beta, \gamma, \dots$$

If we wish to obtain an equation whose roots are

$$F(\alpha), F(\beta), F(\gamma), \dots,$$

then we perform a **transformation of the equation**.

There are four such transformations.

1. Increase each root by k .
2. Decrease each root by k .
3. Multiply each root by k .
4. Take reciprocal of each root.

2. Increase Every Root by k

Suppose

$$f(x) = 0$$

has roots

$$\alpha, \beta, \gamma.$$

We want the new roots

$$\alpha + k, \beta + k, \gamma + k.$$

Method

Replace

$$x$$

by

$$x - k.$$

Therefore

$$\boxed{g(x) = f(x - k).}$$

Why?

Suppose

$$x = \alpha + k.$$

Then

$$x - k = \alpha.$$

Since

$$f(\alpha) = 0,$$

we have

$$f(x - k) = 0.$$

Hence

$$x = \alpha + k$$

is a root.

Example 1

Find the equation whose roots are each greater by 2 than the roots of

$$x^2 - 5x + 6 = 0.$$

Solution

Replace

$$x$$

by

$$x - 2.$$

Then

$$(x - 2)^2 - 5(x - 2) + 6.$$

Expanding,

$$x^2 - 9x + 20.$$

Hence

$$\boxed{x^2 - 9x + 20 = 0.}$$

3. Decrease Every Root by k

Suppose new roots are

$$\alpha - k.$$

Replace

$$x$$

by

$$x + k.$$

Therefore

$$\boxed{g(x) = f(x + k).}$$

Example 2

Find the equation whose roots are each less by 3 than the roots of

$$x^2 - 6x + 8 = 0.$$

Solution

Replace

$$x$$

by

$$x + 3.$$

Then

$$(x + 3)^2 - 6(x + 3) + 8.$$

Simplifying,

$$x^2 + 5.$$

Hence

$$x^2 + 5 = 0.$$

4. Multiply Every Root by k

Suppose new roots are

$$k\alpha, k\beta, \dots$$

Then

$$g(x) = k^n f\left(\frac{x}{k}\right),$$

where

$$n$$

is the degree.

Why multiply by k^n ?

To remove fractions from the coefficients.

Example 3

Find the equation whose roots are twice those of

$$x^2 - 5x + 6 = 0.$$

Solution

Replace

$$x$$

by

$$\frac{x}{2}.$$

Then

$$\left(\frac{x}{2}\right)^2 - 5\left(\frac{x}{2}\right) + 6.$$

Multiply by

$$4.$$

Hence

$$x^2 - 10x + 24 = 0.$$

5. Divide Every Root by k

New roots

$$\frac{\alpha}{k}.$$

Replace

$$x$$

by

$$kx.$$

Example 4

Find the equation whose roots are one-third of the roots of

$$x^2 - 8x + 12 = 0.$$

Solution

Replace

$$x$$

by

$$3x.$$

Then

$$9x^2 - 24x + 12.$$

Therefore

$$\boxed{3x^2 - 8x + 4 = 0.}$$

(after dividing by 3)

6. Reciprocal Roots

Suppose new roots are

$$\frac{1}{\alpha}, \frac{1}{\beta}.$$

Method

Replace

$$x = \frac{1}{y}.$$

Multiply by

$$y^n.$$

Finally replace

$$y$$

by

$$x.$$

The Main Idea

Suppose α is a root of $f(x) = 0$. Then $f(\alpha) = 0$. We want a new equation whose root is $\frac{1}{\alpha}$. Let, $y = \frac{1}{\alpha}$. Then $\alpha = \frac{1}{y}$. Since $f(\alpha) = 0$, substituting $\alpha = \frac{1}{y}$ gives $f\left(\frac{1}{y}\right) = 0$. We multiply by y^n because $f\left(\frac{1}{y}\right) = 0$ gives an equation that is not a polynomial.

Thus, the reciprocal roots are obtained by replacing $x \rightarrow \frac{1}{x}$. This is the key idea.

Formula

If

$$a_n x^n + \dots + a_1 x + a_0 = 0,$$

then reciprocal equation is

$$a_0 x^n + a_1 x^{n-1} + \dots + a_n = 0.$$

The coefficients simply reverse.

Example 5

Find the equation whose roots are the reciprocals of the roots of $2x^3 - 5x^2 + 7x - 4 = 0$.

Solution

Replace x by $1/y$:

$$\frac{2}{y^3} - \frac{5}{y^2} + \frac{7}{y} - 4 = 0.$$

Multiply by y^3 :

$$2 - 5y + 7y^2 - 4y^3 = 0.$$

Rearranging and replacing y by x ,

$$4x^3 - 7x^2 + 5x - 2 = 0.$$

8. Equations from Given Symmetric Functions

Example

Construct a quadratic equation whose roots satisfy $\alpha + \beta = 7$, $\alpha\beta = 12$.

Solution

General equation

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0.$$

Hence

$$\boxed{x^2 - 7x + 12 = 0.}$$

Example

Construct the equation whose roots are

$$\alpha^2, \beta^2,$$

where

$$\alpha, \beta$$

are roots of

$$x^2 - 5x + 6 = 0.$$

Solution

First,

$$\begin{aligned}\alpha + \beta &= 5, \\ \alpha\beta &= 6.\end{aligned}$$

Now

$$\alpha^2 + \beta^2 = 25 - 12 = 13.$$

Also

$$\alpha^2\beta^2 = 36.$$

Hence

$$\boxed{x^2 - 13x + 36 = 0.}$$

Exercise

1. Find the equation whose roots exceed the given roots by 5.

$$x^2 - 7x + 10 = 0.$$

-
2. Find the equation whose roots are twice the roots of

$$x^2 - 4x + 3 = 0.$$

3. Find the equation whose roots are reciprocals of
 $3x^2 - 7x + 2 = 0$.

4. If

$$\alpha, \beta$$

are roots of

$$x^2 - 8x + 12 = 0,$$

find

$$\alpha^3 + \beta^3.$$

5. Find

$$\frac{1}{\alpha} + \frac{1}{\beta}.$$

6. Construct the equation whose roots are
 α^2, β^2 .

Lecture 8: Descartes' Rule of Signs, Upper and Lower Bounds of Roots, and Biquadratic Equations

1. Descartes' Rule of Signs

Definition

Descartes' Rule of Signs gives the **possible number of positive and negative real roots** of a polynomial.

Theorem 1 (Positive Roots)

Let

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0.$$

The number of positive real roots is equal to

- the number of sign changes in

$$P(x),$$

or less than it by an even integer.

Example 1

Determine the possible number of positive roots of

$$P(x) = x^4 - 3x^3 + 5x^2 + 2x - 1.$$

Solution

Signs are

$+, -, +, +, -$

Sign changes

$+\rightarrow -$

one,

$--\rightarrow +$

two,

$+\rightarrow +$

none,

$+\rightarrow -$

three.

Therefore,

there are

3

sign changes.

Hence,

the polynomial has

3 or 1

positive roots.

Theorem 2 (Negative Roots)

Replace

x

by

$-x$.

Then count sign changes.

Example 2

Determine the possible number of negative roots of

$$P(x) = x^4 - 3x^3 + 5x^2 + 2x - 1.$$

Solution

Replace

$$x$$

by

$$-x.$$

Then

$$x^4 + 3x^3 + 5x^2 - 2x - 1.$$

Signs

$$+, +, +, -, -$$

Only one sign change.

Therefore,

Exactly one negative root.

Example 3

Determine the possible positive and negative roots of

$$x^5 + 2x^4 - 7x^3 + x - 5 = 0.$$

Solution

Positive signs

$$+, +, -, +, -$$

Sign changes

$$3.$$

Therefore

Positive roots

$$3$$

or

$$1.$$

For

$$P(-x), \\ -x^5 + 2x^4 + 7x^3 - x - 5.$$

Signs

$$-, +, +, -, -$$

Sign changes

2.

Hence
Negative roots

2

or

0.

2. Upper Bound of Real Roots

Definition

A positive number

k

is called an upper bound if every real root satisfies
 $x \leq k$.

Theorem

If synthetic division by

$x - k$

produces all non-negative numbers,
then

k

is an upper bound.

Example 4

Show that

4

is an upper bound for

$x^3 - 5x + 6$.

Solution

Synthetic division

4	1	0	-5	6
		4	16	44
	1	4	11	50

All entries are non-negative.
Hence

$$\boxed{4}$$

is an upper bound.

Example 5

Find the least positive integer upper bound of

$$x^3 - 7x + 2.$$

Solution

Test

$$2, 3, \dots$$

Synthetic division gives

At

$$3,$$

all coefficients become non-negative.

Hence

$$\boxed{3}$$

is the least integer upper bound.

3. Lower Bound

Definition

A number

$$k$$

is a lower bound if

$$x \geq k$$

for every real root.

Theorem

Use synthetic division with negative numbers.

If the signs alternate, the number is a lower bound.

Example 6

Show that

$$-3$$

is a lower bound of

$$x^3 + 4x^2 - 2x - 8.$$

Solution

Synthetic division

$$\begin{array}{r|rrrrr} -3 & 1 & 4 & -2 & -8 \\ & & -3 & -3 & 15 \\ \hline & 1 & 1 & -5 & 7 \end{array}$$

The signs alternate.

Therefore,

$$\boxed{-3}$$

is a lower bound.

4. Biquadratic Equation**Definition**

A polynomial of the form

$$ax^4 + bx^2 + c = 0$$

is called a **biquadratic equation**.

Method

Put

$$y = x^2.$$

Then

the equation becomes quadratic.

Solve for

$$y,$$

then find

$$x.$$

Example 7

Solve

$$x^4 - 5x^2 + 4 = 0.$$

Solution

Put

$$y = x^2.$$

Then

$$y^2 - 5y + 4 = 0.$$

Hence

$$(y - 1)(y - 4) = 0.$$

Therefore,

$$y = 1$$

or

$$y = 4.$$

Thus

$$x = \pm 1, \pm 2.$$

Hence

$$\boxed{x = \pm 1, \pm 2.}$$

Example 8

Solve

$$2x^4 - 7x^2 + 3 = 0.$$

Solution

Let

$$y = x^2.$$

Then

$$2y^2 - 7y + 3 = 0.$$

Factor,

$$(2y - 1)(y - 3) = 0.$$

Hence

$$y = \frac{1}{2},$$

or

$$3.$$

Therefore,

$$\boxed{x = \pm \frac{1}{\sqrt{2}}, \pm \sqrt{3}.}$$

5. Nature of Roots

Using the discriminant

$$\Delta = b^2 - 4ac.$$

For

$$ax^2 + bx + c = 0,$$

Discriminant Nature of Roots

$\Delta > 0$ Two distinct real roots

$\Delta = 0$ Equal real roots

$\Delta < 0$ Complex conjugate roots

Example 9

Determine the nature of the roots of

$$2x^2 - 4x + 5 = 0.$$

Solution

$$\Delta = 16 - 40 = -24.$$

Hence

Complex conjugate roots.

Exercises

1

Determine the possible number of positive and negative roots of

$$x^4 + 2x^3 - 5x^2 + x - 7 = 0.$$

2

Find the least positive integer upper bound of

$$x^3 - 8x + 6 = 0.$$

3

Show that

$$-4$$

is a lower bound of

$$x^3 + 5x^2 - 6x - 7.$$

4

Solve

$$x^4 - 13x^2 + 36 = 0.$$

5

Solve

$$3x^4 - 10x^2 + 3 = 0.$$

6

Determine the nature of the roots of

$$5x^2 - 8x + 6 = 0.$$